

Korebami O. Adebajo

Mechanical Engineering Graduate Student

Education

- 2021 - 2025 **B.S.E**, Mechanical Engineering, University of South Carolina GPA: **3.9**
Minors in Electrical Engineering and Mathematics
Graduation with Leadership Distinction in Research
- 2025 - 2026 **M.Sc**, Aerospace Engineering, University of South Carolina GPA: **4.0**

Research Experience

February 2022-present **Research Student**, Autonomous Real Time Systems Lab
Department of Mechanical Engineering, **University of South Carolina**

Battery Digital Twin and Electric Aircraft Modeling

- Developed multi-domain MATLAB-Simscape models and digital twins of electric aircraft and lithium-ion battery systems to evaluate battery performance, degradation, and operational feasibility.
- Developed and evaluated physics-informed and semi-empirical battery degradation and Remaining Useful Life (RUL) models using experimental cycling data, realistic flight profiles, SOC, charging rate, and thermal conditions.
- Performed battery characterization and degradation analysis including capacity, Full Equivalent Cycles (FEC), DC internal resistance, open-circuit voltage, and load-dependent voltage measurements.
- Investigated battery replacement costs, energy consumption, and degradation impacts on the profitability of electric aircraft operations.
- Conducted sensitivity and parameter optimization studies to identify degradation-model parameters that provide accurate RUL predictions under realistic aircraft operating conditions.

Cyber-Physical Small Modular Reactor Modeling

- Developed a multi-domain Simscape model of a Small Modular Reactor (SMR) power system including primary and secondary loops, steam generation, thermal transport, and turbine power generation.
- Constructed and evaluated system-level thermal and fluid models for real-time control and grid-interaction studies.
- Analyzed system efficiency, heat transfer, pressure, temperature, and mass-flow behavior and resolved numerical and transient initialization challenges in two-phase fluid simulations.

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Autonomous UAV Sensor Deployment

- Developed ROS-based UAV autonomy pipelines integrating camera processing, path planning, and flight-command nodes for autonomous sensor deployment on civil infrastructure.
- Operated and tested UAV platforms for autonomous localization, tracking, and sensor deployment in GPS-denied environments using vision-based and machine-learning approaches.
- Investigated position and velocity control strategies for GPS-denied UAV operation, including onboard state estimation and altitude drift mitigation.
- Collected flight data and supported development and validation of YOLO-based visual localization and tracking systems for autonomous UAV navigation.

June - **Summer Research Intern**, Therapeutic Technology Design and Development Lab
November - Department of Mechanical Engineering, **Massachusetts Institute of Technology**
2024

- Developed a MATLAB-based predictive model for hydrogen peroxide decomposition in propellant-based actuators for untethered soft robotics.
- Investigated the effects of chamber geometry, catalyst size, reaction rate, temperature, pressure, and residence time on actuator performance.
- Conducted experimental testing and data analysis to validate model predictions and characterize pressurization behavior.
- Redesigned the fuel reactor to achieve a full seal, eliminating leaks and improving fuel retention by 50%.
- Presented research findings at the MIT MSRP Research Symposium and mentored a high school student on a related soft robotics project.

Non-Research Related Experiences

November 2025 - **Ambassador & Application Reviewer**, MIT Summer Research Program

- Guided prospective and participating students through the MIT Summer Research Program and provided insight into the research experience.
- Reviewed and evaluated student applications for participation in the MIT Summer Research Program.

2025 **Guest Speaker**, IEEE-MATLAB Event

- Presented an overview of MATLAB-Simscape applications and multi-domain system modeling to an electrical engineering audience.

January - **Mathematics Grader**, Applied Linear Algebra and Elementary Differential Equations
December - Department of Mathematics, University of South Carolina
2024

- Graded quizzes for approximately 100 students and communicated with instructors regarding grading criteria and student progress.

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- August **Peer Tutor**
 2022 - May 2025 Student Success Centre, University of South Carolina
- Organized weekly academic support sessions for 6–8 students in mathematics, science, introductory engineering, and Creo CAD.
 - Supported over 200 USC students through inclusive academic programming.
- September **Formula 3 EV Tractive Team Lead**, Society of Automobile Engineers
 2023 - May 2024 University of South Carolina
- Implemented a regenerative braking system with low- and high-voltage circuits and instrumentation for the Formula 3 EV race car.
 - Led a team of 7 students in the design and development of the vehicle tractive system.
- September **Skills Robot Build Team Member**, Gamecock Robotics
 2023 - May 2024
- Collaborated on mechanical design and fabrication of VEX competition robots, including integration of piston-powered mechanisms.
- February - **Student Participant**, Close Family Emerging Leaders Program
 April 2024
- Completed leadership development activities focused on collaboration, communication, empathy, and ethical decision-making.
- January - **PwC X SGAP Extern**, Extern
 March 2024
- Conducted comparative analysis to identify opportunities for operational improvement and presented recommendations to NGO officials.
- November **Pi Tau Sigma**, Mechanical Engineering Honor Society
 2022 - Present
- Coordinated event-planning activities and supported initiatives connecting USC engineering students with professional engineers.

Publications and Posters

Peer Reviewed Conference Publications

- [7] **Adebajo, K.O.**, Cooper, N., Mahboub Rad, A., Whetham, M., Shute, N., Huebner, A., Downey, A.R.J., Zhang, J., Khan, J., & Knight, T. “Design Architecture of a Cyber-Physical Small Modular Reactor Emulator for Real-Time Control and Grid Interaction.” ASME IDETC, 2026.
- [6] Burnett, M., Garcia-Sandoval, M.E., **Adebajo, K.O.**, & Downey, A.R.J. “Airframe Architecture and Design of the Swift-UAV: An Open-Source 20 kg-Class 3D-Printed Fixed-Wing Research Platform.” ASME IDETC, 2026.
- [5] Garcia, M., Burnett, M., **Adebajo, K.**, Downey, A., & Allibada, V. “Progress Towards Digital Twin Integration for Online Battery Health Estimation in the Swift-UAV Platform.” AIAA 2026 Regional Student Conferences, 2026. doi:10.2514/6.2026-112511.
- [4] **Adebajo, O. Korebami**, Anthony, G., Peskar, J., Downey, A.R.J., Chen, Y., & Hu, C. “Impact of Charging Rate and State of Charge on Electric Aircraft Battery Degradation Using a Multi-

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Domain Model with Realistic Southeastern U.S. Flight Paths.” AIAA SCITECH 2025 Forum, 2025. doi:10.2514/6.2025-2705.

- [3] Cooper, N., Mahboub Rad, A., **Adebajo, K.**, Downey, A.R.J., & Zhang, J. “Physics-Based Prediction of Remaining Useful Life of Lithium-Ion Batteries in Fuel Cell-Battery Hybrid Aircraft Using an Open-Source Modelica Framework.” IMECE 2025, 2025. doi:10.1115/imece2025-168114.
- [2] Kaushik, H., Mahboub Rad, A., **Adebajo, K.**, Badakhshan, S., Cooper, N., Downey, A., & Zhang, J. “Hybrid Powertrain Optimization for Regional Aircraft Integrating Hydrogen Fuel Cells and Aluminum Air Batteries.” IDETC-CIE 2025, 2025. doi:10.1115/detc2025-168547.
- [1] Gollob, S.D., Comoretto, A., Yu, Q., Fernandez, M., **Adebajo, K.O.**, Overvelde, J.T.B., Becker, K., & Roche, E. “An Actuator-Scale, Pressure-Amplifying, Monopropellant Engine for Untethered Pneumatic Systems.” IEEE RoboSoft Conference, 2025.

Accepted and Under Review

- [2] **Adebajo, K.O.**, Smith, C., Downey, A.R.J., & Queenan, C. “Battery Energy Density Requirements for Economically Viable Short-Range Electric Cargo Flights.” AIAA SCITECH 2027 Forum. **Accepted.**
- [1] Garcia-Sandoval, M.E., Burnett, M., Gardner, M.A., Killedar, D., **Adebajo, K.O.**, Downey, A.R.J., & Ziehl, P. “Uncertainty-Aware Coupled Design–Manufacturing Optimization for High-Rate Additive Manufacturing Using KRISP-U.” AIAA SCITECH 2027 Forum. **Accepted.**

Conference Publications

- [3] Shute, N., Khan, M.A., Satme, J.N., Zhang, M., **Adebajo, K.**, Downey, A.R.J., & Folsom, M. “Vision-Based UAV Localization Using a Fisheye Ground-Camera Network for Sensor Package Deployment in Post-Disaster Structure Inspection.” SPIE 2026. doi:10.1117/12.3095325.
- [2] Khan, M.A., Satme, J.N., Zheng, M., Tadamatla, S., Phillips, E., **Adebajo, K.**, Shute, N., Yount, R., & Downey, A.R.J. “Machine Learning-Based UAV Localization for Under-Structure Sensor Deployment in GPS-Denied Environments.” SPIE 2026. doi:10.1117/12.3089966.
- [1] Zheng, Q., Khan, A., Satme, J.N., **Adebajo, K.**, Yount, R., & Downey, A.R.J. “Stereo YOLO UAV Localization and Tracking Enabling Autonomous Sensor Deployment on Critical Infrastructure.” Submitted to AIAA SCITECH 2026.

In Preparation

- [2] **Adebajo, K.O.** & Downey, A.R.J. “Are Current Semi-Empirical Battery Degradation Models Ready for Electric Aircraft? A Critical Review.” In preparation for *Energies*.
- [1] **Adebajo, K.O.**, Downey, A.R.J., Zhang, J., Cooper, N., Flemming, M., Stanzione, K., & Queenan, C. “Battery Degradation and Replacement Cost Impacts on Profitability in Electric Light Utility Aircraft Using a Multiphysics Modeling Approach.” *Applied Energy*.

Poster Presentations

- [3] **Adebajo, O. Korebami.** “Simulink Modeling of an Electric Airplane.” Discover USC 2023. **Discover USC Award for Excellence and Achievement.**
- [2] **Adebajo, O. Korebami.** “Battery Degradation Prediction Aided by Multi-Domain Modeling.” SC Battery Symposium, 2024.
- [1] **Adebajo, O. Korebami.** “Experimental Analysis of Catalyzed Hydrogen Peroxide Decomposition for Monopropellant-Powered Actuators.” MIT MSRP Research Symposium, 2024.

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Honors and Awards

Honors

- [5] **2026 Data Visualization Award** "Digital Twin Optimization for Battery Degradation in Electric Aircraft"
- [4] **Discover USC Honourable Mention** "Multi-domain Modeling of an Electric Airplane" Poster Award for excellence and achievement 2023
- [3] **Capstone Scholar and Fellow** University of South Carolina, 2021-Present
- [2] **Dean's List** University of South Carolina, GPA > 3.5, Fall 2022 - Present
- [1] **President's List** University of South Carolina, 4.0 GPA list, Fall 2021 and Spring 2022

Scholarships

- [4] **Colorstack Funding** \$1500 academic scholarship awarded in 2024
- [3] **Boeing Scholarship** \$1500 academic scholarship awarded in 2024
- [2] **Herty B. Mechanical Engineering Scholarship** \$1000 academic scholarship awarded in 2023
- [1] **Academic Scholar Distinction Award** \$2000 academic scholarship awarded in 2021

Research Grants

- [4] **Magellan Scholar Award** "Multi-domain Modeling of an Electric Airplane" \$2500 research funding awarded in April 2024
- [3] **Michael A. and Elizabeth Ann Sutton Fellowship** "Multi-domain Modeling of an Electric Airplane" \$2250 research funding awarded in February 2023
- [2] **Magellan Apprentice Grant** "Simulink Modeling of an Electric Airplane" \$1000 research funding awarded in March 2023
- [1] **Magellan Journey Grant** "Simulink Modeling of an Electric Airplane" \$1000 research funding awarded in July 2022